

```
#import numpy as np
```

```
arr = np.array([10, 20, 30, 40])
```

```
print(arr)
```

OUTPUT:

```
[10 20 30 40]
```

```
#import pandas as pd
```

```
# Create DataFrame
```

```
data = {
```

```
    "Name": ["John", "Anna", "Peter", "Lucy"],
```

```
    "Marks": [85, 72, 90, 60]
```

```
}
```

```
df = pd.DataFrame(data)
```

```
# Calculate average marks
```

```
average_marks = df["Marks"].mean()
```

```
print("Average Marks:", average_marks)
```

```
# Add Pass/Fail column (pass if marks >= 70)
```

```
df["Result"] = df["Marks"] >= 70
```

```
# Print DataFrame
```

```
print(df)
```

OUTPUT:

Average Marks: 76.75

	Name	Marks	Result
--	------	-------	--------

0	John	85	True
---	------	----	------

1	Anna	72	True
---	------	----	------

2	Peter	90	True
---	-------	----	------

3	Lucy	60	False
---	------	----	-------

```
# Step 1: Import libraries
import numpy as np
from sklearn.linear_model import LinearRegression

# Step 2: Create data
# x = hours studied, y = marks scored
x = np.array([1, 2, 3, 4, 5]).reshape(-1, 1) # feature (2D)
y = np.array([2, 4, 6, 8, 10])             # target (1D)

# Step 3: Create model and train
model = LinearRegression()
model.fit(x, y)

# Step 4: Make prediction
predicted = model.predict([[6]])
print("Predicted marks for 6 hours:", predicted[0])
```

OUTPUT:

Predicted marks for 6 hours: 12.0

```
# Step 1: Import matplotlib.pyplot as plt
import matplotlib.pyplot as plt

# Step 2: Prepare data
x = [1, 2, 3, 4, 5]
y = [2, 4, 6, 8, 10]

# Step 3: Create plot
plt.plot(x, y)

# Step 4: Add title and labels
plt.title("Simple Line Plot")
plt.xlabel("X-axis")
plt.ylabel("Y-axis")

# Step 5: Show the plot
plt.show()
```

OUTPUT:

